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SCHOOL OF ARTIFICIAL INTELLIGENCE

Generative AI

Syllabus

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Overview

Ready to build production-grade AI? This program equips developers to deploy reliable generative AI solutions. We'll move past theory and focus on the proven implementation patterns you need. You'll master production essentials like model selection, cost estimation, and reliable prompt engineering to build efficient apps. You'll also implement lightweight model adaptation using PEFT. Then, you'll build end-to-end RAG systems, using vector databases to connect LLMs to your data and evaluate quality with frameworks like RAGAs. Finally, you'll dive into advanced multimodal applications that process text, images, and audio. You'll enforce structured outputs with Pydantic and implement system observability to build, trace, and debug modern AI apps.

Nanodegree Program | Intermediate | 56 hours | 4.9 (167 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

Generative AI Fluency | Database fundamentals | Deep learning | Intermediate Python | Hugging Face | Parameter-Efficient Fine-Tuning | Transformer neural networks | PyTorch | Implementing Role-Based Prompting | Foundation Models | Prompt Engineering | Basic Python

You will also need to be able to communicate fluently and professionally **in written and spoken English**.

Skills You'll Learn

Generative AI Fundamentals

Foundation Model Concepts | Generative AI Fluency | Generative AI Evaluation Implementation | Ethical AI Awareness | Generative AI Evaluation | Text generation | Basic PyTorch | Prompt Engineering | Foundation Models | Neural network basics | Deep learning | Prompt Engineering for Classification | Foundation Models | Model Adaptation | Applying Generative AI to Use Cases | Generative AI Awareness | Reinforcement Fine-Tuning using GRPO

Large Language Models (LLMs) and Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation | Retrieval-Augmented Generation | Large Language Models | LLM Analysis and Capabilities | Prompt Engineering

Multimodal AI Applications

Image pre-processing | Word embeddings | Ethical AI | AI Audio and Speech Analysis | Yolo algorithm | Model evaluation | Text generation | Computer vision fluency | Image classification | Large Language Models | Image generation | Image segmentation | Computer Vision Transformers | Diffusion Models | Gemini API | Tokenization | Multimodal AI Implementation

Courses

01. Generative AI Fundamentals

21 hours

Employ the abilities of Generative AI with a deep dive into fundamentals. This course examines how various models are developed, how they work, and how to use them to their full potential.

L1	Introduction to Generative AI Foundations	Explore core principles, tools, and ethical use of Generative AI, and discover its real-world impact and foundational models powering creative applications.
L2	Generative AI Overview	Explore the fundamentals of generative AI, its key modalities, advanced capabilities, and essential ethical considerations shaping responsible AI development.
L3	Accessing OpenAI API Keys	
L4	Applications of Generative AI	Explore real-world applications of Generative AI, including LLM-assisted coding, and learn to prompt, validate, and improve AI-generated code and tests.
L5	Introduction to Foundation Models	Discover foundation models: large, versatile AI systems trained on massive datasets that generalize across tasks, surpassing traditional models in scalability and adaptability.
L6	Building Applications using Foundation Models	Learn to build text classifiers with foundation models, using zero-shot and few-shot prompt engineering for tasks like sentiment and spam detection, and evaluate classifier accuracy.
L7	How Generative AI Works	Learn how generative AI creates new data with architectures like Transformers and diffusion models, and how training enables creativity, reasoning, and task-specific abilities.
L8	Evaluating Generative AI Models	Learn how to assess generative AI using human evaluation, exact metrics, AI judges, and benchmarks, ensuring robust performance for open-ended, probabilistic model outputs.
L9	Implementing Evaluations for Generative AI Models	Learn practical techniques to evaluate generative AI models, from Exact Match to ROUGE, semantic similarity, code correctness, Pass@k, and LLM-as-a-Judge scoring.
L10	Neural Networks and Multilayer Perceptrons	Explore neural networks from perceptrons to multilayer perceptrons, learning how they adapt via training, gradient descent, and backpropagation to solve complex AI tasks.

L11	Implementing Neural Networks using Pytorch	Learn to implement neural networks in PyTorch by mastering tensors, model building, loss functions, optimizers, data loading, and complete training loops for practical machine learning.
L12	Model Interpretability and Ethics	Explore AI model interpretability and ethics, including bias, misinformation, environmental impact, and fairness for responsible development and deployment of AI technologies.
L13	Generating Text using LLMs	Discover how LLMs generate text token by token using Hugging Face's Transformers, from tokenization to model use, and explore hands-on demos with efficient generation methods.
L14	Role-Based Prompting	Explains the theory of using roles or personas to control the tone, style, and expertise of an LLM's output.
L15	Implementing Role-Based Prompting with Python	Provides hands-on practice in iteratively developing a role-based prompt to create a believable historical figure persona.
L16	Adapting Foundation Models	Learn to adapt foundation models for specialized tasks using prompt engineering, RAG, fine-tuning, model compression, and agentic AI tools for efficient, tailored AI solutions.
L17	Applying PEFT on Foundation Models	Learn to efficiently customize foundation models with PEFT and SFT, using LoRA to teach LLMs new skills like spelling via hands-on data preparation and fine-tuning.
L18	Post-Training Foundation Models	Explore post-training for foundation models, including supervised and preference fine-tuning, to align AI with human values, improve usability, and ensure responsible interactions.
L19	Reinforcement Fine-tuning on Foundation Models	Learn to fine-tune LLMs for structured tasks like counting and spelling using GRPO and LoRA, applying reinforcement-based reward functions for targeted skill improvements.
L20	Project: Teaching an LLM to Count!	Teaching an LLM to count the number of letters in a word using GRPO.

02. Large Language Models (LLMs) and Retrieval Augmented Generation (RAG)

17 hours

Master Large Language Models (LLMs) and build sophisticated text generation applications in this hands-on course. You'll master prompt engineering techniques, optimize model selection and costs, and dive deep into Retrieval-Augmented Generation (RAG), using vector databases to ground AI responses in external data and eliminate hallucinations. Finally, you'll evaluate system performance with RAGAS and showcase your skills by building an end-to-end RAG application.

L1	Introduction to LLMs and Retrieval Augmented Generation (RAG)	Introduces Large Language Models (LLMs), their core concepts, and the course structure. Covers prerequisites, environment setup, and defines Retrieval-Augmented Generation (RAG).
L2	The Large Language Model Landscape	Explore the four core capabilities of LLMs: generation, summarization, classification, and reasoning. Real-world applications and the importance of RAG for building trust.
L3	Implementing a Chatbot with an LLM	Learn to build a stateful chatbot using an LLM. Covers managing conversation history, using system prompts to define behavior, and understanding message roles (system, user, assistant).
L4	LLM Prompting & Inference	Defines prompt engineering and its components. Explains how to control LLM outputs using inference parameters like temperature, top-p, max tokens, and stop sequences.
L5	Applied Prompting and Inference	Apply prompting techniques hands-on. Implement Chain of Thought (CoT) prompting to improve reasoning and test how different inference parameters change model behavior.
L6	Prompt Instruction Refinement	Explains the theory of systematically refining prompt instructions by modifying components like Role, Task, Context, Examples, and Output Format.
L7	Applying Prompt Instruction Refinement with Python	Provides hands-on practice iteratively refining a prompt to transform a generic recipe analyzer into a precise dietary consultant that produces structured JSON.
L8	Tokens, Embeddings, and Vector Search	Covers the foundations of NLP for LLMs. Defines tokenization, embeddings as semantic vectors, and vector search (similarity search) as the basis for finding relevant information.
L9	Implementing Tokens, Embeddings, and Vector Search	Hands-on practice with tokenization. Implementing embedding generation and vector search to build a semantic search system from scratch.
L10	Strategic Model Selection & Economics	Learn the business trade-offs of model selection. Covers performance, cost, speed, and control (TCO). Compares general-purpose (generation) vs. specialized (reasoning) models.
L11	Applying Model Selection and Economics	Apply model selection theory. Calculate Total Cost of Ownership (TCO) including error costs. Implement a hybrid model routing system to balance cost and quality.
L12	Retrieval Augmented Generation (RAG) Workflow	Introduces the RAG architecture. Compares naive vs. advanced modular RAG. Covers the data ingestion pipeline, focusing on data formats and intelligent chunking strategies.
L13	Semantic Search with Vector Databases for RAG	Explains semantic search and the role of vector databases. Covers indexing algorithms (HNSW) for speed and advanced retrieval techniques like HyDE and re-ranking.

L14	Prompt Engineering for RAG Synthesis	Learn to write prompts for RAG. Covers grounding answers in context, handling conflicts, managing uncertainty, and enforcing verifiability by generating inline citations.
L15	Implementing RAG with Vector Databases	Build a complete RAG system. Practice vector database operations in ChromaDB, including adding documents, applying metadata filters, and implementing a retrieval and generation pipeline.
L16	Evaluating RAG Systems	Learn to evaluate RAG system quality. Introduces key metrics: Context Precision, Context Recall, Faithfulness, and Answer Relevancy. Covers frameworks like RAGAS.
L17	RAG Evaluation Implementation	Implement a RAG evaluation pipeline using the RAGAS framework. Learn to calculate and interpret quality metrics to diagnose and improve your RAG system's performance.
L18	Project: Project: NASA Mission Intelligence: Developing a RAG-Based Chat System	Build an end-to-end RAG chatbot. Ingest NASA mission data, build a vector search pipeline, generate answers, and evaluate the system's quality.

03. Multimodal AI Applications

18 hours

Learn how computers process and understand image data, then harness the power of the latest Generative AI models to create new images.

L1	Introduction	
L2	Multimodal AI Fundamentals	Discover multimodal AI fundamentals and technologies, including models and use cases that process and generate text, images, audio, and video for richer, real-world applications.
L3	Using Multimodal AI Technologies	Explore practical applications of multimodal AI by using APIs and open-source models for image captioning and audio transcription, with hands-on exercises and secure credential handling.
L4	Transformers & Multimodal Processing	Explore how transformers unify text, images, audio, and video through attention, embeddings, and fusion strategies, powering state-of-the-art multimodal understanding and generation.
L5	Multimodal AI Tooling	Explore practical tools for building multimodal AI apps, compare commercial and open-source options, and use Pydantic AI to create reliable, structured, vendor-agnostic workflows.

L6	Introduction to Enterprise Visual Content Processing	Explore enterprise visual content processing: core computer vision tasks, digital image representation, and real-world applications for efficiency, safety, and automation.
L7	Vision Pre-processing Pipelines with HuggingFace	Explore vision data pipelines using HuggingFace, from dataset loading to resizing and normalization, with demos and hands-on exercises for effective image pre-processing.
L8	Understanding Embeddings in Computer Vision	Learn how embeddings convert images into compact vectors for efficient search, enable cross-modal tasks with models like CLIP, and power large-scale, robust computer vision systems.
L9	Image Search Using CLIP Embeddings	Explore how to build text-to-image and image-to-image search using CLIP embeddings, combining theory, real-world demos, hands-on practice, and solution walkthroughs.
L10	Using Multimodal Model APIs for Vision	Explore multimodal vision APIs: prompt design, parameter tuning, structured outputs, cost control, integration, and best practices for robust, efficient image analysis.
L11	Gemini Vision API Basics	Explore Gemini Vision API basics by practicing image moderation, learning to analyze images and implement moderation workflows using real-world examples and guided hands-on exercises.
L12	Vision Transformer Models & Architectures	Explore Vision Transformer models: core architecture, image tokenization, self- and cross-attention, and top models (SAM, RT-DETR, DINOv2) for segmentation, detection, and enterprise use.
L13	Using Vision Transformers	Explore vision transformers with hands-on demos: extract image embeddings using DINOv2 and perform object detection and segmentation using RT-DETR and SAM2.1 models.
L14	Vision-Language Models	Learn how vision-language models align images and text for tasks like search, captioning, and VQA, with focus on architectures, applications, data needs, and deploying for enterprise use.
L15	Multimodal Vision Applications with CLIP	Explore zero-shot image classification and auto-labeling for driving scenes using CLIP, enabling efficient, scalable multimodal vision applications.
L16	Diffusion Models & Image Generation	Explore how diffusion models generate images by reversing noise through iterative denoising, inspired by physical diffusion processes and key to modern generative AI developments.
L17	Introduction to Enterprise Audio Processing	Discover enterprise audio processing, core speech tasks (transcription, diarization, sentiment, TTS), key use cases, and strategies for value and integration in modern businesses.
L18	Audio Data Representation	Explore how audio is digitized for AI: sample rate, bit depth, channels, formats, and mel spectrograms for speech, plus challenges and best practices in audio preprocessing and analysis.

L19	Audio Processing with librosa	Explore audio processing with librosa: load, resample, convert, and analyze audio files; visualize with mel spectrograms and apply techniques through hands-on exercises.
L20	Sound Retrieval and Classification	Explore audio embeddings for efficient sound classification and retrieval, using models like CLAP to enable semantic search and robust text-based audio analysis at scale.
L21	Sound Retrieval and Classification with CLAP	Explore using CLAP for sound retrieval, similarity, and zero-shot classification, then apply these skills to detect fan on/off states in real audio data.
L22	Speech Processing	Discover automatic speech recognition with Whisper: a robust, multilingual, open-source model for accurate transcription, translation, and speech processing in real-world audio.
L23	Implementing Speech Processing with Whisper & Gemini	Explore real-world speech transcription and translation with Whisper and Gemini, using Python to process, segment, and align audio with text, including multilingual support.
L24	Audio Intelligence	Explore advances in Audio Intelligence: multimodal systems, speech recognition, TTS, enterprise controls, creative workflows, and ethics for robust, secure, and accessible audio solutions.
L25	Audio Sentiment Analysis with Gemini	Explore audio sentiment and command analysis using Pydantic AI and Gemini; learn to extract emotions and recognize spoken commands from audio with real-world datasets and hands-on exercises.
L26	Audio Classification and Moderation	Explore voice content moderation: real-time and batch pipelines, compliance, privacy, layered detection, and operational excellence for secure and fair audio classification.
L27	Building a Basic Voice Moderation System with Gemini	Learn to build a voice moderation system using Gemini to transcribe audio, detect personal data disclosures, and flag policy violations in customer service recordings.
L28	Introduction to Enterprise Video Processing	Discover how enterprise video AI overcomes temporal complexity using smart frame selection for efficient understanding, search, classification, moderation, and generation at scale.
L29	AI Models for Video Understanding	Explore key AI models like YOLO for real-time detection, CoTracker and TimeSformer for motion and temporal understanding, enabling advanced, scalable enterprise video analytics.
L30	Implementing Object Recognition & Tracking	Learn how to detect and track objects in videos using YOLOv9, apply multi-object tracking, handle small objects, and count items crossing boundaries in practical scenarios.
L31	Video Understanding & Search	Explore methods for video analysis and search using foundation models and CLIP4Clip, balancing temporal understanding, cost, and retrieval accuracy for enterprise applications.

L32	Video Understanding & Search with Gemini & Clip4Clip	Explore video understanding with Gemini and Clip4Clip: learn automated video description, key moment detection, and natural language video search using AI models and structured outputs.
L33	Video Classification & Moderation	Learn to classify and moderate video by modeling temporal patterns, handling real-world challenges, and combining automation with human oversight for scale, accuracy, and compliance.
L34	Video Classification & Moderation with Gemini	Learn to build automated systems for video classification and moderation with Gemini and Pydantic AI, including action recognition and safety compliance in real-world scenarios.
L35	Video Generation	Explore generative video AI tools and workflows that turn text, images, or footage into dynamic content for marketing, training, and creative use while ensuring quality and compliance.
L36	Video Generation with Veo 3	Learn to generate marketing videos with Veo 3 using both text-to-video and image-to-video workflows, and understand their strengths, limitations, and real-world applications.
L37	Multimodal AI Deployment	Explore deployment of multimodal AI systems for text, images, audio, video via unified APIs, multi-API orchestration, and custom solutions, balancing speed, cost, and control.
L38	Implementation Tools and Serving Strategies	Explore tools and strategies for implementing, serving, and monitoring AI solutions, from rapid prototyping to production, including unified APIs, orchestration, and managed platforms.
L39	Using Gradio and Pydantic AI	Learn to build multimodal chatbots and analysis apps using Gradio and Pydantic AI, covering async programming, media inputs, rate limiting, and interface customization.
L40	Multimodal AI Performance Monitoring and Logging	Learn to monitor and log multimodal AI systems, tracking performance, costs, and failures across modalities for optimized, reliable, and coherent production deployments.
L41	Logging and Performance Monitoring with Gradio and Arize Phoenix	Learn to implement logging and performance monitoring for multimodal AI chatbots using Gradio and Arize Phoenix, enabling robust analytics, debugging, and cost tracking.
L42	Evaluating Multimodal Applications	Learn how to evaluate multimodal AI apps using user feedback systems and testing methods, blending human review, automated metrics, and continuous monitoring for quality improvement.
L43	Testing Multimodal Apps with Pydantic AI Evals	Learn to build robust testing frameworks for multimodal AI apps using Pydantic Evals, covering structured outputs, semantic evaluation, custom evaluators, and hands-on exercises.

L44	Scaling Multimodal AI Architecture	Learn strategies to scale multimodal AI: unified APIs, multi-API pipelines, and custom deployments, focusing on performance, cost, reliability, and architectural trade-offs.
L45	Project: OmniTrainer: Multimodal Customer Service Trainer	In this project, students will create an AI agent that simulates customer service scenarios and specialized monitoring agents that analyze communications across text, images, videos, and audio.

Meet Your Instructors



Brian Cruz

Head of AI Engineering, Advocate

Brian Cruz is the Head of Data and AI Engineering at Advocate, where generative AI is used to help disabled individuals access the government benefits they deserve. Previously he led Machine Learning Engineering for Einstein Guidance at Salesforce, and he also established the Center of Excellence for AI and ML at Samba TV. He studied Mathematics at UC Berkeley.



Eduardo Mota

Sr. Cloud Data Architect

Eduardo Mota is an AWS Cloud Architect and Machine Learning Specialist. He has achieved numerous cloud certifications including Machine Learning and Data Analytics. His experience at DoiT and AWS showcases his expertise in cloud architecture and optimization and solidifies his position as a trusted expert in cloud innovation and AI.



Giacomo Vianello

Director, Machine Learning Engineer

Giacomo Vianello is Director, Machine Learning Engineer at Moody's, where he leads AI innovation and deploys ML solutions across insurance and real estate. He's passionate about demystifying advanced ML concepts and teaching engineers how to build production-grade AI systems that actually ship.

Why Udacity



Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately



24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices